Unit-III

PARTIAL DIFFERENTIAL EQUATIONS

Topic Learning Objectives:

After completion of this unit, students will be able to

(i) Form PDE by the elimination of arbitrary constants and arbitrary functions.

(ii) Solve the Lagrange's linear equations.

(iii) Apply the method of separation of variables to solve I and II order PDE.

(iv) Find the solution of heat, wave and Laplace equation.

Introduction

Partial Differential Equations (often referred to as PDEs) frequently arise in the study or formulation of fundamental laws of nature and in the mathematical analysis of wide variety of problems in applied mathematics, mathematical physics and engineering sciences. Historically PDEs originated from the study of surfaces in geometry and for solving wide variety of problems in mechanics. Many physical problems in science and engineering, when formulated mathematically, give rise to PDEs. In order to understand the behavior of the mathematical model it is necessary to have some knowledge about the mathematical character, properties and the solution of the governing PDE. Many physical processes fundamental to science and engineering are governed by PDEs, that is, equations involving partial derivatives. The most familiar of these processes are heat conduction and wave propagation.

An equation involving partial derivatives of a (unknown) function u or z that depends on two or more independent variables is called a PDE.

The most important PDEs are the wave equations that can model the vibrating string and the vibrating membrane, the heat equation for temperature in a bar or wire and the Laplace equation for electrostatic potentials. PDEs are very important in dynamics, elasticity, heat transfer, electromagnetic theory and quantum mechanics. They have a much wider range of applications than ODEs, which can model only the simplest physical systems.

A PDE for a function $u(x, y, \ldots)$ is a relationship between u and $x, y, \ldots$ and its partial derivatives $u_x, u_y, u_{yy}, u_{xx}, u_{xy}, \ldots$ and can be expressed as

$$F(x, y, \ldots, u, u_x, u_y, u_{yy}, u_{xx}, u_{xy}, \ldots) = 0. \tag{1}$$

where F is some function of all or some of its arguments.

An order of a PDE is the order of the highest order derivatives in the PDE.

The general PDE of first order is of the form

$$F(x, y, \ldots, u, u_x, u_y) = 0. \tag{2}$$

The general PDE of second order is of the form

$$F(x, y, \ldots, u, u_x, u_y, u_{xx}, u_{xy}, u_{yy}) = 0. \tag{3}$$

Following notations are used in the future course of study.
$u_x = p, u_y = q, u_{xx} = r, u_{xy} = s, u_{yy} = t.$

Formation of PDE:

Partial differential equation can be formed by two methods

(i) By the elimination of arbitrary constants,

(ii) By the elimination of arbitrary functions

from a given relation involving three or more variables.

(I) Formation of PDE by the elimination of arbitrary constants

Example 1. *Form the PDE by eliminating the arbitrary constants a and b from the relation* $z = (x - a)^2 + (y - b)^2$.

Solution: Given

$$z = (x - a)^2 + (y - b)^2 \tag{4}$$

Differentiating the above equation partially w.r.t. 'x', we get

$$\frac{\partial z}{\partial x} = 2(x - a) \Rightarrow p = 2(x - a) \tag{5}$$

Now differentiating the given equation partially w.r.t. 'y', we get

$$\frac{\partial z}{\partial y} = 2(y - b) \Rightarrow q = 2(y - b) \tag{6}$$

Now eliminate the arbitrary constants a and b from the above equations. Substituting the equations (5) and (6) in the given equation (4), we get

$$z = \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2 \Rightarrow 4z = p^2 + q^2$$

is the required first order PDE.

Example 2. *Form the PDE by eliminating the arbitrary constants a and b from the relation* $z = xy + y\sqrt{x^2 - a^2} + b$.

Solution: Given

$$z = xy + y\sqrt{x^2 - a^2} + b \tag{7}$$

Differentiating the above equation partially w.r.t. 'x' and 'y', we get

$$\frac{\partial z}{\partial x} = y + y\frac{1}{2\sqrt{x^2-a^2}}2x \Rightarrow p = y + xy\frac{1}{\sqrt{x^2-a^2}} \tag{8}$$

$$\frac{\partial z}{\partial y} = x + \sqrt{x^2-a^2} \Rightarrow q - x = \sqrt{x^2-a^2} \tag{9}$$

substituting the equation (9) in (8), we get

$$p = y + xy\frac{1}{q-x} \Rightarrow (p-y)(q-x) = xy$$

is the required first order PDE.

Example 3. *Form the PDE by eliminating the arbitrary constants a, b and c from the relation $z = ax + by + cxy$.*

Solution: Given

$$z = ax + by + cxy \tag{10}$$

Differentiating the above equation partially w.r.t. 'x' and 'y', we get

$$\frac{\partial z}{\partial x} = a + cy \Rightarrow p = a + cy \tag{11}$$

$$\frac{\partial z}{\partial y} = b + cx \Rightarrow q = b + cx. \tag{12}$$

It is not possible to eliminate the arbitrary constants from the above three equations.
$\therefore$ differentiate equation(11) partially w.r.t. 'y',

$$\frac{\partial}{\partial y}[\frac{\partial z}{\partial x}] = c \Rightarrow \frac{\partial^2 z}{\partial y \partial x} = c \tag{13}$$

substituting the equation (13) in equations (11) and (12), we get

$$p = a + sy \Rightarrow a = p - sy \tag{14}$$

$$q = b + sx \Rightarrow b = q - sx \tag{15}$$

Now substituting the equations (13), (14) and (15) in the given equation (10), we get

$$z = (p - sy)x + (q - sx)y + sxy \Rightarrow z = px + py - sxy$$

is the required 2nd order PDE.

Exercise:

(i) Form a PDE which represents a family of spheres of all whose radius is k (parameter) and whose centers lie in the $xy-$plane. Hint: $(x-a)^2 + (y-b)^2 + z^2 = k^2$.

(ii) Form the PDE representing all planes that are at a constant perpendicular distance d from the origin. Hint: $lx + my + nz = p, l^2 + m^2 + n^2 = 1$.

(iii) Form the PDE by eliminating the arbitrary constants a and b from the relation $z = ax + by + ab$.

(iv) Form a PDE by eliminating th arbitrary constants a, b and c from the relation $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Answers:

(i) $z^2(p^2 + q^2 + 1) = k^2$.

(ii) $Mpx + Mqy - M^2z = p$, where $M = \sqrt{\frac{1}{1 + (\frac{\partial z}{\partial x})^2 + (\frac{\partial z}{\partial y})^2}}$.

(iii) $z = px + qy + pq$.

(iv) $zr + p^2 - \frac{z}{x}p = 0$.

(II) Formation of PDE by elimination of arbitrary functions from a given relation involving three or more variables

Example 4. *Form a PDE by eliminating the arbitrary function from the relation* $z = f(x^2 - y^2)$.

Solution: Given

$$z = f(x^2 - y^2) \tag{16}$$

We need to eliminate the function f, differentiating the above equation partially w.r.t. 'x', we get

$$\frac{\partial z}{\partial x} = f'(x^2 - y^2) \times (2x - 0) \Rightarrow p = 2xf'(x^2 - y^2). \tag{17}$$

Now differentiating the given equation partially w.r.t, 'y', we get

$$\frac{\partial z}{\partial y} = f'(x^2 - y^2) \times (0 - 2y) \Rightarrow q = -2yf'(x^2 - y^2). \tag{18}$$

Dividing the equation (17) by (18), we get

$$\frac{p}{q} = \frac{2xf'(x^2 - y^2)}{-2yf'(x^2 - y^2)} \Rightarrow \frac{p}{q} = \frac{-x}{y} \Rightarrow xq + yp = 0$$

is the required first order PDE.

Example 5. *Form a PDE by eliminating the arbitrary function ϕ from the relation $z = e^{my}\phi(x - y)$, where m is a constant.*

Solution: Given

$$z = e^{my}\phi(x-y) \tag{19}$$

differentiating the above equation partially w.r.t. 'x' and 'y', we get

$$\frac{\partial z}{\partial x} = e^{my}\phi'(x-y) \Rightarrow p = e^{my}\phi'(x-y) \tag{20}$$

$$\frac{\partial z}{\partial y} = e^{my}\phi'(x-y)(0-1) + e^{my}(m)\phi(x-y)$$

$$\Rightarrow q = -e^{my}\phi'(x-y) + me^{my}\phi(x-y) \tag{21}$$

using equations (19) and (20) in the above equation (21), we get

$$q = -p + mz \Rightarrow p + q = mz$$

is the required first order PDE.

Example 6. *Form a PDE by eliminating the arbitrary functions f and g from the relation* $u = f(x+at) + g(x-at)$.

Solution: Given

$$u = f(x+at) + g(x-at) \tag{22}$$

differentiating the above equation partially w.r.t. 'x' and 't', we get

$$\frac{\partial u}{\partial x} = f'(x+at) + g'(x-at) \tag{23}$$

$$\frac{\partial u}{\partial t} = f'(x+at)(a) + g'(x-at)(-a) \tag{24}$$

differentiating the above equations (23) and (24) partially w.r.t. 'x' and 't' respectively, we get

$$\frac{\partial^2 u}{\partial x^2} = f''(x+at) + g''(x-at) \tag{25}$$

$$\frac{\partial^2 u}{\partial t^2} = a^2 f''(x+at) + a^2 g''(x-at) = a^2\left(f''(x+at) + g''(x-at)\right) \tag{26}$$

using equation (25) in the above equation (26), we get

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

is the required second order PDE.

Example 7. *Form a PDE by eliminating the arbitrary function f from the relation* $f(z/x^3,\ y/x) = 0$.

Solution: Given

$$f\left(\frac{z}{x^3}, \frac{y}{x}\right) = 0 \tag{27}$$

Let $u = \frac{z}{x^3}$ and $v = \frac{y}{x}$, then

$$\Rightarrow f\left(\frac{z}{x^3}, \frac{y}{x}\right) = f(u, v) = 0 \tag{28}$$

Now differentiating the above equation w.r.t x and y, we get

$$\frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = 0 \tag{29}$$

$$\frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} = 0 \tag{30}$$

Substituting, $\frac{\partial u}{\partial x} = \frac{xp - 3z}{x^4}$, $\frac{\partial u}{\partial y} = \frac{q}{x^3}$, $\frac{\partial v}{\partial x} = \frac{-y}{x^2}$, $\frac{\partial v}{\partial y} = \frac{1}{x}$ in the above equation, we get

$$\frac{\partial f}{\partial u}\left(\frac{xp - 3z}{x^4}\right) + \frac{\partial f}{\partial v}\left(\frac{-y}{x^2}\right) = 0 \tag{31}$$

$$\frac{\partial f}{\partial u}\left(\frac{q}{x^3}\right) + \frac{\partial f}{\partial v}\left(\frac{1}{x}\right) = 0 \tag{32}$$

Now eliminating $\frac{\partial f}{\partial u}$ and $\frac{\partial f}{\partial v}$ from above equations, i.e.,

$$\begin{vmatrix} \frac{xp - 3z}{x^4} & \frac{-y}{x^2} \\ \frac{q}{x^3} & \frac{1}{x} \end{vmatrix} = 0$$

$$\Rightarrow xp - yq = 3z \tag{33}$$

which is the required PDE.

Exercise: Form the PDE by eliminating arbitrary functions for the following

(i) $\phi(x^2 + 2yz,\ y^2 + 2zx) = 0$

(ii) $f(x^2 + y^2,\ z - xy) = 0$

(iii) $z = y^2 + 2f(1/x + \log y)$

(iv) $z = f(\frac{xy}{z})$

Answers:

(i) $(xz - y^2)p + (yz - x^2)q = (xy - z^2)$.

(ii) $yp - xq = y^2 - x^2$,

(iii) $qy = 2y^2 - px^2$.

(iv) $xp = yq$.

Solution of PDE

Solution of Lagrange's linear equation

A partial differential equation of the first order is called linear if it is of the first degree in p and q. The first order linear partial differential equation of the form $Pp + Qq = R$, where P, Q and R are functions of x, y, z is known as Lagrange's Linear equation.

Working Rule to solve the equation $Pp + Qq = R$

1 Write the equation in the form $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$ which is known as **subsidiary** or **auxiliary** equations.

2 Find two independent integrals of these subsidiary equations say $u = c_1$ and $v = c_2$ then the general solution is given by $f(u, v) = 0$ or $u = f(v)$ where f is arbitrary function.

Example 8. *Solve* $xp + yq = z$.

Solution: Given $xp + yq = z$ which is of the form $Pp + Qq = R$.
Here $P = x$, $Q = y$, $R = z$.
The subsidiary equation is given by

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$
$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$$

Taking 1st and 2nd ratios

$$\frac{dx}{x} = \frac{dy}{y}$$

Integrating,

$$\log x = \log y + \log c_1$$
$$\log x - \log y = \log c_1$$
$$\log \frac{x}{y} = \log c_1$$
$$\frac{x}{y} = c_1$$

Taking 2nd and 3rd ratios

$$\frac{dy}{y} = \frac{dz}{z}$$

Integrating,

$$\log y = \log z + \log c_2$$
$$\log y - \log z = \log c_2$$
$$\log \frac{y}{z} = \log c_2$$
$$\frac{y}{z} = c_2$$

$\therefore$ the solution of the given PDE is given by $f(c_1, c_2) = 0 \Rightarrow f(x/y, y/z) = 0$.

Example 9. *Solve* $\frac{y^2z}{x}p + xzq = y^2$.

Solution: Given $\frac{y^2z}{x}p + xzq = y^2$ which is of the form $Pp + Qq = R$.

Here $P = \frac{y^2z}{x}$, $Q = xz$, $R = y^2$.

The subsidiary equation is given by

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{\frac{y^2z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2}$$

Taking 1st and 2nd ratios

$$\frac{x\,dx}{y^2z} = \frac{dy}{xz}$$

$$\Rightarrow x^2dx = y^2dy$$

Integrating,

$$\frac{x^3}{3} = \frac{y^3}{3} + c$$

$$\Rightarrow x^3 - y^3 = c_1.$$

Taking 1st and 3rd ratios

$$\frac{x\,dx}{y^2z} = \frac{dz}{y^2}$$

$$\Rightarrow xdx = zdz$$

Integrating,

$$\frac{x^2}{2} = \frac{z^2}{2} + c$$

$$\Rightarrow x^2 - z^2 = c_2.$$

$\therefore$ the solution of the given PDE is given by

$f(c_1, c_2) = 0 \Rightarrow f(x^3 - y^3, x^2 - z^2) = 0.$

Example 10. *Solve* $(z - y)p + (x - z)q = (y - x)$.

Solution: Given $(z - y)p + (x - z)q = (y - x)$ which is of the form $Pp + Qq = R$.

Here $P = z - y$, $Q = x - z$, $R = y - z$.

The subsidiary equation is given by

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{z-y} = \frac{dy}{x-z} = \frac{dz}{y-z}$$

Choosing $1,\ 1,\ 1$ as the multipliers, we get each ratio

$$= \frac{dx + dy + dz}{z - y + x - z + y - x} = 0$$

$$\Rightarrow dx + dy + dz = 0$$

Integrating,

$$x + y + z = c_1.$$

Choosing $x,\ y,\ z$ as the multipliers, we get each ratio

$$= \frac{xdx + ydy + zdz}{xz - xy + yx - yz + zy - zx} = 0$$

$$\Rightarrow xdx + ydy + zdz = 0$$

Integrating,

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c$$

$$\Rightarrow x^2 + y^2 + z^2 = c_2.$$

$\therefore$ the solution of the given PDE is given by

$f(c_1, c_2) = 0 \Rightarrow f(x + y + z, x^2 + y^2 + z^2) = 0.$

Example 11. *Solve* $(z^2 - 2yz - y^2)p + (xy + zxz)q = xy - zx.$

Solution: Given $(z^2-2yz-y^2)p+(xy+zx)q = xy-zx$ which is of the form $Pp+Qq = R.$
Here $P = z^2 - 2yz - y^2$, $Q = xy + zx$, $R = xy - zx.$
The subsidiary equation is given by

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$
$$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + zx} = \frac{dz}{xy - zx}$$

Taking 2nd and 3rd ratios

$$\frac{dy}{x(y+z)} = \frac{dz}{x(y-z)}$$
$$\Rightarrow (y-z)dy = (y+z)dz$$
$$\Rightarrow ydy - zdy = ydz + zdz$$
$$\Rightarrow ydy - zdy - ydz - zdz = 0$$
$$\Rightarrow ydy - d(yz) - zdz = 0$$

Integrating,

$$\frac{y^2}{2} - yz - \frac{z^2}{2} = c$$
$$\Rightarrow y^2 - 2yz - z^2 = c_1.$$

Choosing $x,\ y,\ z$ as the multipliers, we get each ratio

$$= \frac{xdx + ydy + zdz}{xz^2 - 2xyz - xy^2 + xy^2 + xyz + xyz - z^2x} = 0$$
$$\Rightarrow xdx + ydy + zdz = 0$$

Integrating,

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c$$
$$\Rightarrow x^2 + y^2 + z^2 = c_2.$$

$\therefore$ the solution of the given PDE is given by
$f(c_1, c_2) = 0 \Rightarrow f(y^2 - 2yz - z^2, x^2 + y^2 + z^2) = 0.$

Exercise: Solve the following PDE's

(i) $(mz - ny)p + (nx - lz)q = ly - mx$

(ii) $xzp + yzq = xy$

(iii) $(y^2 + z^2)p - x(yq - z) = 0$

Answers:

(i) $f(x^2 + y^2 + z^2, lx + my + nz) = 0.$

(ii) $f(\frac{x}{y}, xy - z^2) = 0,$

(iii) $f(\frac{y}{z}, x^2 + y^2 + z^2) = 0.$

Solution of partial differential equations by the method of separation of variables

Separation of variables is a powerful technique to solve PDE.
For a PDE in the function u of two independent variables x and y, assume a solution in the form

$$u(x,\, y) = X(x)Y)(y), \tag{34}$$

where $X(x)$ is a function of x alone and $Y(y)$ is a function of y alone. Then substituting u and its derivatives, reduces the PDE to the form

$$f(X, X', X'', ...) = g(Y, Y', Y'', ...), \tag{35}$$

which is separable in X and Y.

Since LHS of equation (35) is a function of x alone and RHS of (35) is a function of y alone, then equation (35) must be equal to a common constant say k.
Thus the equation (35) reduces to

$$f(X, X', X'', ...) = k, \tag{36}$$
$$g(Y, Y', Y'', ...) = k. \tag{37}$$

Thus the determination of solution to PDE reduces to the determination of solution to the ODE and solving equations (36) and (37) gives the solution of the given PDE.

Example 12. *Solve $u_{xx} - u_y = 0$ by the method of separation of variables.*

Solution: Given PDE is

$$u_{xx} - u_y = 0 \tag{38}$$

Assume the solution of the given PDE (38) to be of the form

$$u(x,\, y) = X(x)Y(y) \tag{39}$$

differentiating the equation (39) w.r.t. 'x' and 'y', we get

$$u_x = Y\frac{dX}{dx}\ ,\ u_{xx} = \frac{d}{dx}(u_x) = Y\frac{d^2X}{dx^2} \text{ and } u_y = X\frac{dY}{dy} \tag{40}$$
$$\therefore (38) \Rightarrow Y\frac{d^2X}{dx^2} - X\frac{dY}{dy} = 0$$
$$\Rightarrow Y\frac{d^2X}{dx^2} = X\frac{dY}{dy}.$$

Separating the variables X and Y, we get

$$\frac{1}{X}\frac{d^2X}{dx^2} = \frac{1}{Y}\frac{dY}{dy} = k, \text{ a constant (say)} \tag{41}$$

considering

$$\frac{1}{X}\frac{d^2X}{dx^2} = k \tag{42}$$

$$\Rightarrow \frac{d^2X}{dx^2} = kX$$

$$\Rightarrow \frac{d^2X}{dx^2} - kX = 0 \tag{43}$$

which is a 1st order homogeneous linear differential equation (LDE) with constant coefficients, to solve this we consider the auxiliary equation which is given by

$$m^2 - k = 0 \Rightarrow m = \pm\sqrt{k}.$$

$\therefore$ the solution of the LDE (43) is given by

$$X(x) = c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x}.$$

Now consider

$$\frac{1}{Y}\frac{dY}{dy} = k \Rightarrow \frac{dY}{Y} = kdy \quad \text{(by separating the variables)}$$

on integration,

$$\log Y = ky + \log c_3 \Rightarrow \log Y - \log c = ky \Rightarrow \log\frac{Y}{c_3} = ky \Rightarrow Y(y) = c_3 e^{ky} \tag{44}$$

$\therefore$ now by substituting both $X(x)$ and $Y(y)$ in the assumed $u(x,\, y) = X(x)Y(y)$, we get

$$u(x,\, y) = \left(c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x}\right) c_3 e^{ky}$$

which is the required solution of the given PDE.

Example 13. *Solve by the method of separation of variables* $\dfrac{\partial u}{\partial x} = 2\dfrac{\partial u}{\partial t} + u$ *given that* $u(x, 0) = 6e^{-3x}$.

Solution: Given PDE is

$$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u. \tag{45}$$

Assume the solution of the given PDE (45) to be of the form

$$u(x,\, t) = X(x)T(t). \tag{46}$$

Differentiating the equation (46) w.r.t. 'x' and 't', we get

$$u_x = T\frac{dX}{dx} \text{ and } u_t = X\frac{dT}{dt} \tag{47}$$

$$\therefore (45) \Rightarrow T\frac{dX}{dx} = 2X\frac{dT}{dt} + XT = X\left(2\frac{dT}{dt} + T\right) \tag{48}$$

separating the variables

$$\Rightarrow \frac{1}{X}\frac{dX}{dx} = \frac{2}{T}\frac{dT}{dt} + 1 = -p^2 \text{ (say)} \tag{49}$$

considering
$\frac{1}{X}\frac{dX}{dx} = -p^2$
separating the variables
$\Rightarrow \frac{dX}{x} = -p^2 dx$
on integration
$\Rightarrow \log X = -p^2 x + \log c_1$
$\Rightarrow X = c_1 e^{-p^2 x}$

Now consider
$\frac{2}{T}\frac{dT}{dt} + 1 = -p^2$
$\frac{dT}{T} = -\frac{(p^2+1)}{2}dt$ (by separating the variables)
on integrating, we get
$\log T = -\frac{(p^2+1)}{2}t + \log c_2$
$\Rightarrow T = c_2 e^{-\frac{(p^2+1)}{2}t}$

now by substituting both $X(x)$ and $T(t)$ in the assumed $u(x,\, t) = X(x)T(t)$, we get

$$u(x,\, t) = c_1 e^{-p^2 x} c_2 e^{-\frac{(p^2+1)}{2}t} = c_1 c_2 e^{-p^2 x - \frac{(p^2+1)}{2}t}. \tag{50}$$

Next, in order to find the values of the arbitrary constants c_1 and c_2, making use of the given initial condition $u(x, 0) = 6e^{-3x}$,
i.e., putting $t = 0$ in the equation (50), we get
$6e^{-3x} = c_1 c_2 e^{-p^2 x} \Rightarrow c_1 c_2 = 6$ and $p^2 = 3$.
Substituting these values in the equation (50), we get

$$u(x,\, t) = 6e^{-3x-2t} \tag{51}$$

which is the required solution of the given PDE.

Exercise: Solve the following by the method of separation of variables

(i) $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$ and $u(0, y) = e^{-5y}$.

(ii) $u_{xt} = e^{-t}cos(x)$ with $u(x, 0) = 0$ and $\frac{\partial u}{\partial t}(0, t) = 0$

Answers:

(i) $u(x, y) = e^{2x-5y}$.

(ii) $u(x,t) = sinx - e^{-t}sinx.$

Solution of Wave equation in one dimension by the method of separation of variables

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \tag{52}$$

where $c^2 = \dfrac{T}{\rho}$ the phase speed, T is the tension, and ρ density of the string.

Solution: Assume the separable solution to the given PDE (52) in the form

$$u(x,\, t) = X(x)T(t) \tag{53}$$

differentiating the equation (53) twice both w.r.t. 'x' and 't', we get

$$u_x = T\frac{dX}{dx},\ u_{xx} = T\frac{d^2X}{dx^2},\ u_t = X\frac{dT}{dt} \text{ and } u_{tt} = X\frac{d^2T}{dt^2} \tag{54}$$

$$\therefore (52) \Rightarrow X\frac{d^2T}{dt^2} = c^2T\frac{d^2X}{dx^2}.$$

Separating the variables

$$\Rightarrow \frac{1}{c^2T}\frac{d^2T}{dt^2} = \frac{1}{X}\frac{d^2X}{dx^2} = k \text{ (say)}. \tag{55}$$

Here k can be 0, $+ve$, $-ve$, we need to find the solution for all the three cases.

Case i: When $k = 0$

From eqution (55) we have

$$\frac{d^2T}{dt^2} = 0$$

On integration,we get

$$\frac{dT}{dt} = c_1$$

Once again on integration,we get

$$T(t) = c_1t + c_2.$$

From eqution (55) we have

$$\frac{d^2X}{dx^2} = 0$$

On integration,we get

$$\frac{dX}{dx} = c_3$$

Once again on integration,we get

$$X(x) = c_3x + c_4.$$

Substituting these values in the equation (53), we get

$$u(x,\, t) = (c_3 + c_4x)\,(c_1 + c_2t) \tag{56}$$

which is the required solution of the given PDE.

Case ii: When k is $+ve$, let $k = p^2$

From eqution (55) we have	From eqution (55) we have
$(D^2 - c^2p^2)T = 0$	$(D^2 - p^2)X = 0$
Then the A.E. is given by	Then the A.E. is given by
$m^2 - c^2p^2 = 0 \Rightarrow m = \pm cp$	$m^2 - p^2 = 0 \Rightarrow m = \pm p$
$\therefore$ the solution is given by	$\therefore$ the solution is given by
$T(t) = c_5e^{cpt} + c_6e^{-cpt}$	$X(x) = c_7e^{px} + c_8e^{-px}$

Substituting these values in the equation (53), we get

$$u(x,\, t) = \left(c_7e^{px} + c_8e^{-px}\right)\left(c_5e^{cpt} + c_6e^{-cpt}\right) \tag{57}$$

which is the required solution of the given PDE.

Case iii: When k is $-ve$, let $k = -p^2$

	From eqution (55) we have		From eqution (55) we have
	$(D^2 + c^2p^2)T = 0$		$(D^2 + p^2)X = 0$
	Then the A.E. is given by		Then the A.E. is given by
	$m^2 + c^2p^2 = 0 \Rightarrow m = \pm icp$		$m^2 + p^2 = 0 \Rightarrow m = \pm ip$
$\Rightarrow$	$T(t) = c_9\cos(cpt) + c_{10}\sin(cpt)$	$\Rightarrow$	$X(x) = c_{11}\cos(px) + c_{12}\sin(px)$

Substituting these values in the equation (53), we get

$$u(x,\, t) = \left(c_9\cos(cpt) + c_{10}\sin(cpt)\right)\left(c_{11}\cos(px) + c_{12}\sin(px)\right) \tag{58}$$

which is the required solution of the given PDE.

Of these three solutions, we have to choose the solution which is consistent with the physical nature of the problem. As we will be dealing with problems on vibrations, u must be a periodic function of x and t. Accordingly the solution given in case(iii) is the only best suitable solution of the wave equation. i.e.

$$u(x,\, t) = \left(c_{11}\cos(px) + c_{12}\sin(px)\right)\left(c_9\cos(cpt) + c_{10}\sin(cpt)\right). \tag{59}$$

Solution of Heat equation in one dimension by the method of separation of variables

$$\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}, \tag{60}$$

where $c^2 = \dfrac{K}{s\rho}$ the thermal diffusivity, K thermal conductivity, s specific heat, and ρ density of the material of the body.

Solution: Assume the solution of the given PDE (60) to be of the form

$$u(x,\, t) = X(x)T(t) \tag{61}$$

differentiating the equation (61) w.r.t. 'x' and 't', we get

$$u_x = T\frac{dX}{dx},\ u_{xx} = T\frac{d^2X}{dx^2} \text{ and } u_t = X\frac{dT}{dt} \tag{62}$$
$$\therefore (60) \Rightarrow X\frac{dT}{dt} = c^2T\frac{d^2X}{dx^2}$$

separating the variables

$$\Rightarrow \frac{1}{X}\frac{d^2X}{dx^2} = \frac{1}{c^2T}\frac{dT}{dt} = k \text{ (say)} \tag{63}$$

Here k can be 0, $+ve$, $-ve$, we need to find the solution for all the three cases.

Case i: When $k = 0$

From eqution (63) we have
$\frac{dT}{dt} = 0$
On integration,we get
$T(t) = c_1.$

From eqution (63) we have
$\frac{d^2X}{dx^2} = 0$
On integration,we get
$\frac{dX}{dx} = c_2$
Once again on integration,we get
$X(x) = c_2x + c_3.$

Substituting these values in the equation (61), we get

$$u(x,\, t) = c_1\,(c_2x + c_3) \tag{64}$$

which is the required solution of the given PDE.

Case ii: When $k = +ve$, let $k = p^2$

From eqution (63) we have
$(D - c^2p^2)T = 0$
Then the A.E. is given by
$m - c^2p^2 = 0 \Rightarrow m = c^2p^2$
$\therefore$ the solution is given by
$T(t) = c_4e^{c^2p^2t}$

From eqution (63) we have
$D^2X - p^2X = 0$
Then the A.E. is given by
$m^2 - p^2 = 0 \Rightarrow m = \pm p$
$\therefore$ the solution is given by
$X(x) = c_5e^{px} + c_6e^{-px}$

Substituting these values in the equation (61), we get

$$u(x,\, t) = c_4e^{c^2p^2t}\left(c_5e^{px} + c_6e^{-px}\right) \tag{65}$$

which is the required solution of the given PDE.

Case iii: When $k = -ve$, let $k = -p^2$

From eqution (63) we have	From eqution (63) we have
$(D + c^2p^2)T = 0$	$D^2X + p^2X = 0$
Then the A.E. is given by	Then the A.E. is given by
$m + c^2p^2 = 0 \Rightarrow m = -c^2p^2$	$m^2 + p^2 = 0 \Rightarrow m = \pm pi$
$\therefore$ the solution is given by	$\therefore$ the solution is given by
$T(t) = c_7e^{-c^2p^2t}$	$X(x) = c_8\cos(px) + c_9\sin(px)$

Substituting these values in the equation (61), we get

$$u(x,\, t) = (c_8\cos(px) + c_9\sin(px))\, c_7e^{-c^2p^2t} \tag{66}$$

which is the required solution of the given PDE.

Of these three solutions, we have to choose the solution which is consistent with the physical nature of the problem. As we will be dealing with problems on heat conduction, accordingly the solution given in case(iii) is the only best suitable solution of the heat equation. i.e.

$$u(x,\, t) = (c_8\cos(px) + c_9\sin(px))\, c_7e^{-c^2p^2t}. \tag{67}$$

Solution of Laplace equation in two dimension by the method of separation of variables

$$\frac{\partial^2u}{\partial x^2} + \frac{\partial^2u}{\partial y^2} = 0 \tag{68}$$

Solution: Assume the solution of the given PDE (68) to be of the form

$$u(x,\, y) = X(x)Y(y) \tag{69}$$

differentiating the equation (69) w.r.t. 'x' and 'y' twice, we get

$$u_x = Y\frac{dX}{dx},\ u_{xx} = Y\frac{d^2X}{dx^2},\ u_y = X\frac{dY}{dy} \text{ and } u_{yy} = X\frac{d^2Y}{dy^2} \tag{70}$$

$$\therefore (68) \Rightarrow Y\frac{d^2X}{dx^2} + X\frac{d^2Y}{dy^2} = 0$$

separating the variables

$$\Rightarrow \frac{1}{X}\frac{d^2X}{dx^2} = -\frac{1}{Y}\frac{d^2Y}{dy^2} = k \text{ (say)} \tag{71}$$

Here k can be 0, $+ve$, $-ve$, we need to find the solution for all the three cases.

Case i: When $k = 0$

From eqution (71) we have, $\frac{d^2X}{dx^2} = 0$

On integration,we get

$\frac{dX}{dx} = c_1$

Once again on integration,we get

$X(x) = c_1x + c_2$

From eqution (71) we have, $\frac{d^2Y}{dy^2} = 0$

On integration,we get

$\frac{dY}{dy} = c_3$

Once again on integration,we get

$X(x) = c_3y + c_4$

Substituting these values in the equation (69), we get

$$u(x,\, y) = (c_1x + c_2)\,(c_3y + c_4) \tag{72}$$

which is the required solution of the given PDE.

Case ii: When $k = +ve$, let $k = p^2$

From eqution (71) we have

$D^2X - p^2X = 0$

Then the A.E. is given by

$m^2 - p^2 = 0 \Rightarrow m = \pm p$

$\therefore$ the solution is given by

$X(x) = c_5e^{px} + c_6e^{-px}$

From eqution (71) we have

$D^2Y + p^2Y = 0$

Then the A.E. is given by

$m^2 + p^2 = 0 \Rightarrow m = \pm pi$

$\therefore$ the solution is given by

$X(x) = c_7\cos(py) + c_8\sin(py)$

Substituting these values in the equation (69), we get

$$u(x,\, y) = \left(c_5e^{px} + c_6e^{-px}\right)(c_7\cos(py) + c_8\sin(py)) \tag{73}$$

which is the required solution of the given PDE.

Case iii: When $k = -ve$, let $k = -p^2$

From eqution (71) we have

$D^2X + p^2X = 0$

Then the A.E. is given by

$m^2 + p^2 = 0 \Rightarrow m = \pm pi$

$\therefore$ the solution is given by

$X(x) = c_9\cos(px) + c_{10}\sin(px)$

From eqution (71) we have

$D^2Y - p^2Y = 0$

Then the A.E. is given by

$m^2 - p^2 = 0 \Rightarrow m = \pm p$

$\therefore$ the solution is given by

$Y(y) = c_{11}e^{py} + c_{12}e^{-py}$

Substituting these values in the equation (69), we get

$$u(x,\, y) = (c_9\cos(px) + c_{10}\sin(px))\left(c_{11}e^{py} + c_{12}e^{-py}\right) \tag{74}$$

which is the required solution of the given PDE.

Of these three solutions, we have to choose that solution which is consistent with the

given boundary conditions. If boundary line on which u is specified to be non-zero is parallel to the x-axis (y constant) then the solution of the form given by case (iii) is used. If boundary line on which u is specified to be non-zero is parallel to the y-axis (x constant) then the solution of the form given by case (ii) is used.